

D860

General Physics II with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A kitchen-science demo at a community center with a new induction cooktop heats 2.0 kg of water from 20 °C to 70 °C. Water's specific heat is 4,186 J/(kg·°C). The pot is stainless steel with a copper base. How much heat did the water absorb?

- A. 209,300 J
- B. 586,040 J
- C. 418,600 J
- D. 753,480 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

5,000 J of heat is added to 0.50 kg of aluminum ($c = 900 \text{ J}/(\text{kg}\cdot^\circ\text{C})$). By how much does its temperature rise?

- A. It cannot be found without the starting temperature.
- B. About 11.1 °C
- C. About 5.6 °C
- D. 0.09 °C

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

On a winter night in a cabin heated by a wood stove, a camper feels warmth on her face from the stove 2 m away even though the air between them feels cool. The stove was built in 1985 and weighs 180 kg. Which mode of heat transfer mainly warms her face?

- A. Convection, since warm air always rises straight to her face.
- B. Radiation, but only because air molecules carry it.
- C. Radiation (infrared) from the hot stove.
- D. Conduction through the air, since the air touches both.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A power plant at a riverside site that employs 250 people makes steam at 527 °C and rejects heat to river water at 27 °C. Its turbines were overhauled last year. What is its maximum (Carnot) efficiency?

- A. 37.5%
- B. 62.5%
- C. About 94.9%
- D. Nearly 100%, since the turbines were just overhauled.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A heat engine's hot reservoir is at 600 K, and its Carnot efficiency is 40%. What is the temperature of its cold reservoir?

- A. 360 K
- B. 360 °C
- C. 1,000 K
- D. 240 K

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fiber-optic installer for a regional internet provider explains that light stays trapped inside a glass core ($n = 1.50$) surrounded by cladding ($n = 1.46$). The cable is buried 1 m deep. What condition allows total internal reflection?

- A. Light hitting the boundary at less than the critical angle, nearly head-on.
- B. Light going from the higher-index core toward the lower-index cladding and hitting the boundary beyond the critical angle.
- C. Light going from the lower-index cladding into the higher-index core.
- D. Light hitting the boundary at any angle at all.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Light passes from water ($n = 1.33$) into glass ($n = 1.50$). Can total internal reflection happen at this boundary?

- A.** Yes, because glass is a good reflector.
- B.** Yes, if the angle exceeds the critical angle.
- C.** Yes, above 48.8° , the water-air critical angle.
- D.** No; light entering a higher-index medium bends toward the normal and cannot totally reflect.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A biology teacher uses a handheld magnifier (a converging lens with a 10 cm focal length) that came in a donated science kit to look at an insect held 6 cm from the lens. The kit also contains tweezers and slides. What kind of image is seen?

- A.** Real, inverted, and magnified.
- B.** Real, inverted, and reduced.
- C.** Virtual, upright, and magnified.
- D.** No image forms.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In a darkened lecture hall, a soap film ($n = 1.33$) in air is lit with 532 nm green laser light in a demo whose laser cost \$40. What is the minimum film thickness that gives a bright reflection?

- A.** 100 nm
- B.** 133 nm
- C.** 177 nm
- D.** 200 nm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An astronomy club at a community college, which meets monthly, views a hydrogen discharge tube through a diffraction grating and sees a red line and a blue-violet line. The tube runs at 5,000 V. Compared with the red line, what produces the blue-violet line?

- A.** An electron dropping across a larger energy gap, giving a higher-energy, shorter-wavelength photon.
- B.** An electron jumping up by absorbing a photon.
- C.** The high tube voltage directly creating blue photons.
- D.** An electron dropping across a smaller energy gap.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hospital's nuclear medicine unit, which serves three counties, receives 80 mg of a tracer isotope with a half-life of 6 hours. The shipment arrived by courier at 7 a.m. How much remains 24 hours later?

- A.** 10 mg
- B.** 20 mg
- C.** None; it is fully gone after two half-lives.
- D.** 5 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A sample of a radioactive isotope drops from 64 g to 8 g in 15 days. What is its half-life?

- A.** 1.875 days
- B.** 15 days
- C.** 7.5 days
- D.** 5 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A kitchen-science demo ~~at a community center with a new induction cooktop~~ heats 2.0 kg of water from 20 °C to 70 °C. Water's specific heat is 4,186 J/(kg·°C). ~~The pot is stainless steel with a copper base.~~ How much heat did the water absorb?

Target word: heat absorbed **Decides:** $\Delta T = 50\text{ °C}$.

A	Dropped element	Drops the 2.0 kg mass.
B	Wrong stem number	Uses the final temperature (70 °C) as ΔT .
C	Correct	$Q = 2.0 \times 4,186 \times 50 = 418,600\text{ J}$.
D	Inverted operation	Adds the temperatures (90 °C) instead of subtracting.

2. Answer B Concept 1 · transfer

5,000 J of heat is added to 0.50 kg of aluminum ($c = 900\text{ J/(kg·°C)}$). By how much does its temperature rise?

Target word: ΔT from Q **Decides:** $\Delta T = Q \div (mc)$.

A	Adjacent category	ΔT depends on Q, m, and c, not on the starting temperature.
B	Correct	$5,000 \div (0.50 \times 900) = 5,000 \div 450 \approx 11.1\text{ °C}$.
C	Dropped element	Divides by c only and drops the mass.
D	Inverted operation	Divides mc by Q — the formula flipped.

3. Answer C Concept 2

On a winter night in a cabin heated by a wood stove, a camper feels warmth on her face from the stove 2 m away even though the air between them feels cool. ~~The stove was built in 1985 and weighs 180 kg.~~ Which mode of heat transfer mainly warms her face?

Target word: heat transfer mode **Decides:** Warmth arrives across cool air.

A	Absolute language	Warm air rises upward, not sideways across 2 m of cool air.
B	Adjacent category	Radiation needs no medium at all.
C	Correct	Radiation travels through the air without heating it much.
D	Confusion set	Air is a poor conductor, and the air in between is cool.

4. Answer B Concept 3

A power plant ~~at a riverside site that employs 250 people~~ makes steam at 527 °C and rejects heat to river water at 27 °C. ~~Its turbines were overhauled last year.~~ What is its maximum (Carnot) efficiency?

Target word: Carnot efficiency **Decides:** 800 K and 300 K.

A	Inverted operation	Reports $T_{\text{cold}}/T_{\text{hot}}$ instead of 1 minus it.
B	Correct	$1 - 300/800 = 0.625$.
C	Dropped element	Drops the kelvin conversion: $1 - 27/527$.
D	Camouflage	Maintenance cannot beat the Carnot limit set by temperatures.

5. Answer A Concept 3 · transfer

A heat engine's hot reservoir is at 600 K, and its Carnot efficiency is 40%. What is the temperature of its cold reservoir?

Target word: T_{cold} from efficiency **Decides:** $T_{\text{c}} = T_{\text{h}}(1 - \eta)$.

A	Correct	$600 \times (1 - 0.40) = 360 \text{ K}$.
B	Confusion set	Right number, wrong unit — the result is in kelvins.
C	Inverted operation	Divides 600 by 0.60; a cold reservoir cannot be hotter than the hot one.
D	Dropped element	Uses 600×0.40 , dropping the “1 –”.

6. Answer B Concept 4

A fiber-optic installer ~~for a regional internet provider~~ explains that light stays trapped inside a glass core ($n = 1.50$) surrounded by cladding ($n = 1.46$). ~~The cable is buried 1 m deep.~~ What condition allows total internal reflection?

Target word: condition for TIR **Decides:** Core index higher than cladding.

A	Confusion set	Steeper-than-critical angles pass through; TIR needs angles greater than critical.
B	Correct	Both conditions are needed.
C	Inverted operation	TIR happens only going from higher to lower index.
D	Absolute language	Below the critical angle, light refracts out.

7. Answer D Concept 4 · transfer

Light passes from water ($n = 1.33$) into glass ($n = 1.50$). Can total internal reflection happen at this boundary?

Target word: TIR direction check **Decides:** Going into a higher index.

A	Adjacent category	Glass mainly transmits; TIR depends on the index order.
B	Inverted operation	There is no critical angle in this direction.
C	Wrong stem number	That angle applies to water into air, not water into glass.
D	Correct	No critical angle exists going from lower to higher index.

8. Answer C Concept 5

A biology teacher uses a handheld magnifier (a converging lens with a 10 cm focal length) ~~that came in a donated science kit~~ to look at an insect held 6 cm from the lens. ~~The kit also contains tweezers and slides.~~ What kind of image is seen?

Target word: image type **Decides:** Object inside the focal length.

A	Inverted operation	That occurs with the object between f and $2f$.
B	Adjacent category	That occurs with the object beyond $2f$.
C	Correct	$1/d_i = 1/10 - 1/6$ gives $d_i = -15 \text{ cm}$ (virtual); $m = 15/6 = 2.5$.
D	Confusion set	No image forms only when the object is exactly at the focal point.

9. Answer A Concept 6

In a darkened lecture hall, a soap film ($n = 1.33$) in air is lit with 532 nm green laser light in a demo whose laser cost \$40. What is the minimum film thickness that gives a bright reflection?

Target word: minimum thickness for constructive reflection **Decides:** One phase flip; $t = \lambda/(4n)$.

A	Correct	$532 \div (4 \times 1.33) = 532 \div 5.32 = 100 \text{ nm}$.
B	Confusion set	Uses $\lambda/4$ with the vacuum wavelength, not the wavelength in the film.
C	Inverted operation	Multiplies by n instead of dividing.
D	Dropped element	Uses $\lambda/(2n)$, dropping the half-wave phase flip.

10. Answer A Concept 7

An astronomy club at a community college, which meets monthly, views a hydrogen discharge tube through a diffraction grating and sees a red line and a blue-violet line. The tube runs at 5,000 V. Compared with the red line, what produces the blue-violet line?

Target word: transition for the bluer line **Decides:** Bluer = higher photon energy.

A	Correct	$E = hf$; a bigger gap makes a higher-frequency photon.
B	Confusion set	Jumping up is absorption; bright lines are emission.
C	Camouflage	The voltage excites atoms; line colors are set by the atom's energy levels.
D	Inverted operation	A smaller gap gives lower-energy, redder light.

11. Answer D Concept 8

A hospital's nuclear medicine unit, which serves three counties, receives 80 mg of a tracer isotope with a half-life of 6 hours. The shipment arrived by courier at 7 a.m. How much remains 24 hours later?

Target word: amount remaining **Decides:** $24 \text{ h} = 4 \text{ half-lives}$.

A	Wrong stem number	Counts only 3 half-lives.
B	Inverted operation	Divides by the number of half-lives (4) instead of by 2^4 .
C	Absolute language	Each half-life removes half of what is left; it never hits zero that way.
D	Correct	$80 \times (\frac{1}{2})^4 = 80 \div 16 = 5 \text{ mg}$.

12. Answer D Concept 8 • transfer

A sample of a radioactive isotope drops from 64 g to 8 g in 15 days. What is its half-life?

Target word: half-life from decay **Decides:** $64 \rightarrow 32 \rightarrow 16 \rightarrow 8$ is 3 half-lives.

A	Inverted operation	Divides 15 by the ratio 8 instead of by the number of half-lives.
B	Dropped element	Drops the step of counting half-lives.
C	Wrong stem number	Assumes 2 half-lives.
D	Correct	3 half-lives in 15 days = 5 days each.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Specific heat: $Q = mc\Delta T$	Heat absorbed equals mass $\times$ specific heat $\times$ temperature change; ΔT is final minus initial, not either temperature alone.	Plugging in a single temperature for ΔT or dropping the mass.	A mass, a specific heat, and two temperatures.
2	Conduction, convection, radiation	Conduction needs contact, convection needs a moving fluid, and radiation (infrared) crosses space with no medium needed.	Assuming all warmth travels through the air.	Warmth felt across a gap where the air stays cool.
3	Carnot efficiency	Maximum heat-engine efficiency = $1 - T_{\text{cold}}/T_{\text{hot}}$ with temperatures in kelvins.	Using Celsius, or computing $T_{\text{cold}}/T_{\text{hot}}$ and stopping.	Two reservoir temperatures given in $^{\circ}\text{C}$.
4	Total internal reflection	TIR happens only when light goes from a higher to a lower index of refraction and strikes the boundary beyond the critical angle.	Thinking TIR can happen going into a denser medium or at any angle.	Two indices of refraction and a question about trapping light.
5	Converging lens images	An object inside the focal length of a converging lens gives a virtual, upright, magnified image (a magnifier); outside the focal length the image is real and inverted.	Expecting a real, inverted image from a magnifying glass.	Object distance compared with focal length.
6	Thin-film interference	For a film in air, reflection off the top surface flips phase by half a wavelength, so the thinnest bright reflection is $t = \lambda \div (4n)$.	Ignoring the phase flip ($t = \lambda/2n$) or using the vacuum wavelength in the film.	A film index, a wavelength, and "minimum thickness."
7	Emission lines and photon energy	An emission line comes from an electron dropping between levels; a larger energy gap gives a higher-energy, higher-frequency, shorter-wavelength photon ($E = hf$).	Linking bluer light with a smaller gap or with absorption.	Two emission lines of different colors from one element.
8	Half-life	After n half-lives, the fraction remaining is $(\frac{1}{2})^n$; the amount never simply hits zero after two half-lives.	Dividing by the number of half-lives or assuming linear decay.	A starting amount, a half-life, and an elapsed time.

Course scope (public description): General Physics II with Lab builds on the foundational concepts covered in General Physics I, guiding students through introductory topics in heat and thermodynamics, geometric optics, and the interactions and transformations of energy within atomic and subatomic systems. Students will explore the distinction between heat and temperature, methods of heat transfer, specific heat capacity, the laws of thermodynamics, kinetic theory of gases, and the principles of heat engines and efficiency. The course delves into electromagnetic waves, Snell's law, total internal reflection, thin film interference, ray tracing for lenses and mirrors, dispersion, polarization, diffraction, and the functioning of optical instruments. Finally, students will analyze Interaction and Transformations of Energy, explore atomic models and structures, atomic spectral absorption and emission, radioactivity and radioactive decay, nuclear processes like fusion and fission, and

mass-energy relationships. Throughout the course, students will develop scientific literacy and the ability to apply scientific principles, enhancing their understanding of physical science concepts and their applications.